

Parth K Dubal

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Education

Purdue University

Aug 2024 – Present

B.S. in Aeronautical and Astronautical Engineering, Minor in Business Economics

West Lafayette, IN

- GPA: 3.81 / 4 | Dean's List (4x), Semester Honors (4x)
- Relevant Coursework: Structures and Materials, Thermodynamics, Engineering Design, CAD Modeling in NX, Programming in C/Python/MATLAB

Research

Undergraduate Researcher – PURT-UAVs: Biomedical Narcan Delivery

Aug 2024 – Sept 2025

Purdue University

West Lafayette, IN

- Designed a UAV capable of delivering Narcan during opioid overdose emergencies.
- Modeled structural frameworks using **SolidWorks** and **Fusion 360**, improving top speed by **33%** (45 → 60 mph).
- Awarded 1st Place Data Stewardship Award at the 2025 Spring Undergraduate Research Conference.

Experience

Aerospace Engineering Intern – TarkX Defence Systems

May 2026 – Aug 2026

Project under the Indian Navy

Bangalore, India

- Developed an analytical thermal model of a diffuser nozzle using Python, verifying results against **ANSYS Thermal** to ensure structural integrity under high-temperature gradients.
- Conducted a comprehensive literature review to down-select the optimal **Thrust Vector Control (TVC)** configuration adhering to rigid space, mass, and operational constraints.
- Designed the complete TVC mechanical assembly and executed transient **ANSYS CFD simulations** to evaluate true thrust vectoring performance, off-axis side forces, and fluid-structure interactions.
- Compiled a detailed **Bill of Materials (BOM)** by calculating requirements, sourcing, and sizing flight-grade actuators, high-torque motors, and structural hardware.

Engineer – Purdue Undergraduate Research Propulsion Lab (PURPL)

Aug 2025 – Present

Purdue University

West Lafayette, IN

- **Modular Heatsink Rocket Engine Development (Jan 2026 – Present)**
 - * Developing a modular heatsink rocket engine architecture focusing on thermal management and structural integrity.
 - * Running **NASA CEA** for combustion performance prediction and mixture optimization.
 - * Conducting material selection based on thermo-structural properties at high temperatures, and manufacturability.
 - * Engineered a **1-D transient thermal solver** using Python to determine maximum burn duration by modeling heat flux and material failure thresholds across various propellant chemistries, O/F ratios, and chamber pressures.
- **Secondary Injection Thrust Vectoring Control (Aug 2025 – Dec 2025)**
 - * Designed and analyzed SITVC integration into an in-house rocket engine.
 - * Researched correlations between injection angles, port locations, thrust, and angle of deflection
 - * Created a **python script** to relate and plot effective thrust to injection angles, and angle of deflection

Apprentice Engineer – Materials and Manufacturing Research Laboratories

Dec 2025 – Present

Purdue University

West Lafayette, IN

- Trained on SOPs for Powder and Wire Feed Directed Energy Deposition systems and 5-axis CNC machines.
- Assisted in machine setup, calibration, and parameter selection for additive and subtractive manufacturing processes.
- Expanded working knowledge of equipment capabilities, operability constraints, and post-processing workflows.

Career Lead – Aeronautical & Astronautical Engineering Student Council

Aug 2025 – Present

AAESAC - Purdue University

West Lafayette, IN

- Organized networking events connecting student body of **1,000+** AAE students with over **7+ aerospace companies**.
- Supported the execution of 2 career events by managing on-site logistics and technical requirements for recruiters from major firms like Lockheed Martin, Firefly Aerospace and United Airlines.

Technical Skills

CAD: SOLIDWORKS, NX, Fusion 360

Simulations and Analysis: ANSYS, XFLR5

Programming: MATLAB, Python, C, SQL, HTML/CSS/JS

Interests

Basketball — Sketch & Spray-Paint Art